

Extra credit problems

Math 427

0. Find a mistake or misprint in the book. (The score depends on the type of mistake.)

1. Describe all the motions of the Manhattan plane.

2. Construct a metric space $\mathcal{X}$ and a distance-preserving map $f: \mathcal{X} \rightarrow \mathcal{X}$ that is not a motion of $\mathcal{X}$.

3. Note that the following quantity

$$\tilde{\angle}ABC = \begin{cases} \pi & \text{if } \angle ABC = \pi, \\ -\angle ABC & \text{if } \angle ABC < \pi. \end{cases}$$

can serve as the angle measure; that is, the axioms hold if one changes everywhere $\angle$ to $\tilde{\angle}$.

Show that $\angle$ and $\tilde{\angle}$ are the only possible angle measures on the plane, but without Axiom IIIc, this is not longer true.

4. Show that a composition of three reflections in the sides of a nondegenerate triangle does not have a fixed point.

5. Lines ℓ and m are tangent to two circles of radiuses r and R on such a way the circles are on one side of ℓ and on different sides of m . Let A and B be tangential points of ℓ and Q be the point of intersection ℓ and m . Show that

$$QA \cdot QB = R \cdot r.$$